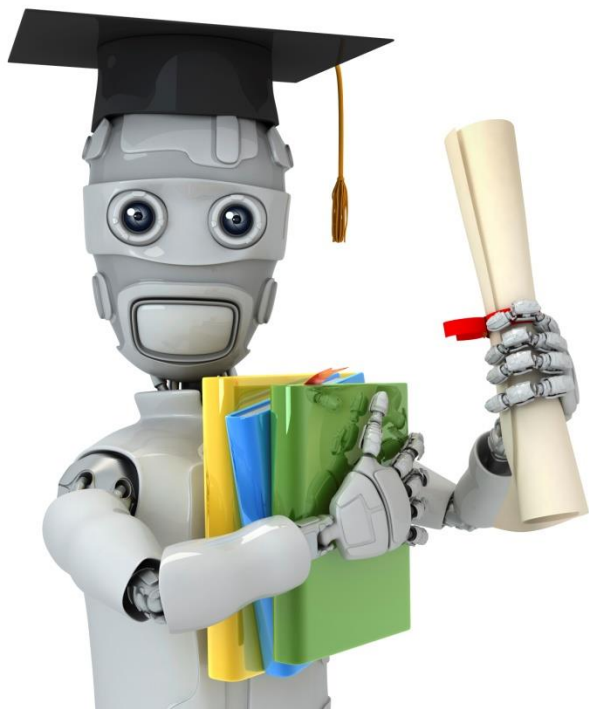
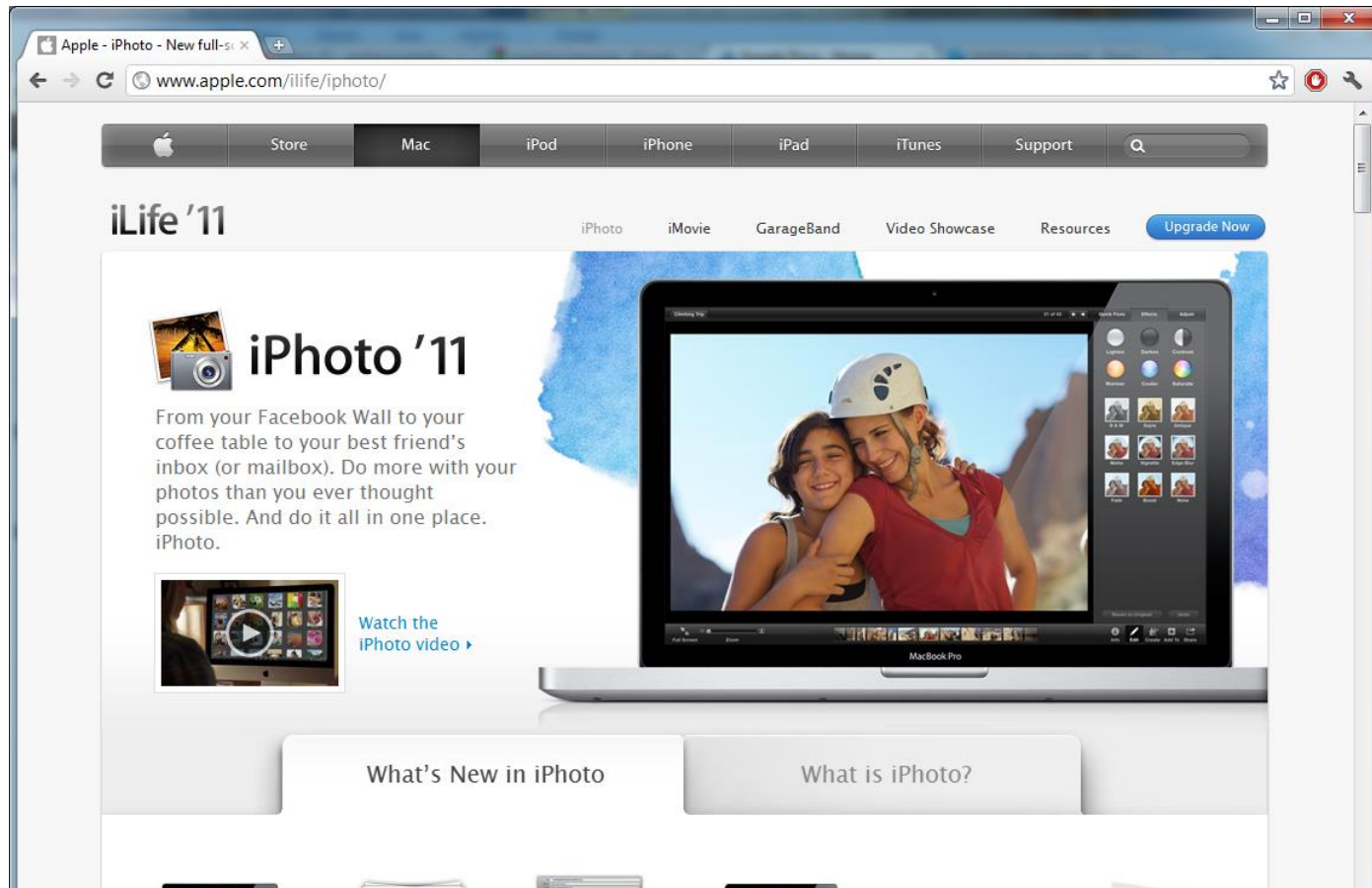




Machine Learning

Introduction

Welcome





Machine Learning

- Grew out of work in AI
- New capability for computers

Examples:

- Database mining
 - Large datasets from growth of automation/web.
 - E.g., Web click data, medical records, biology, engineering
- Applications can't program by hand.
 - E.g., Autonomous helicopter, handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.

Machine

- Gre
- Nev

Example

- Dat
- L
- E
- App



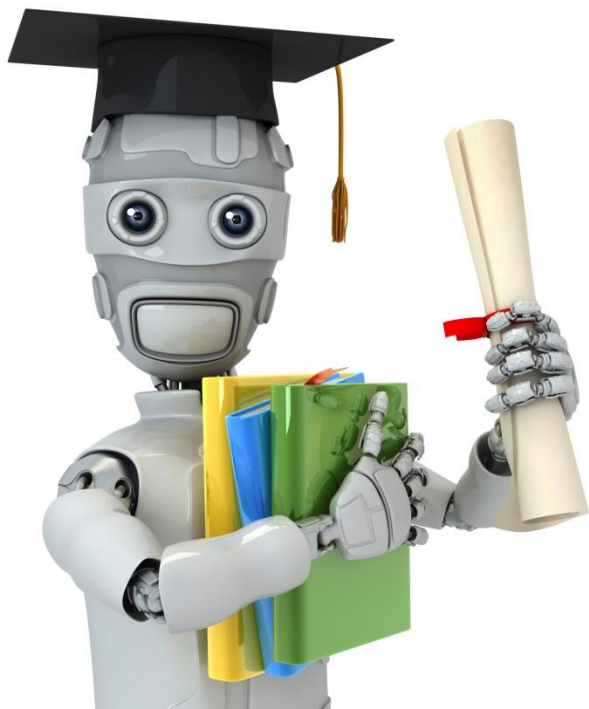
E.g., Automatic document, handwriting recognition, most of

Natural Language Processing (NLP), Computer Vision.

- Self-customizing programs

E.g., Amazon, Netflix product recommendations

- Understanding human learning (brain, real AI).



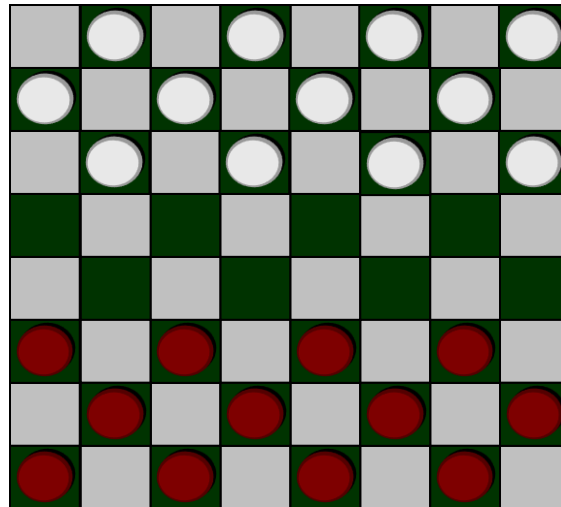
Machine Learning

Introduction

What is machine learning

Machine Learning definition

- Arthur Samuel (1959). Machine Learning:
Field of study that gives computers the ability
to learn without being explicitly programmed.



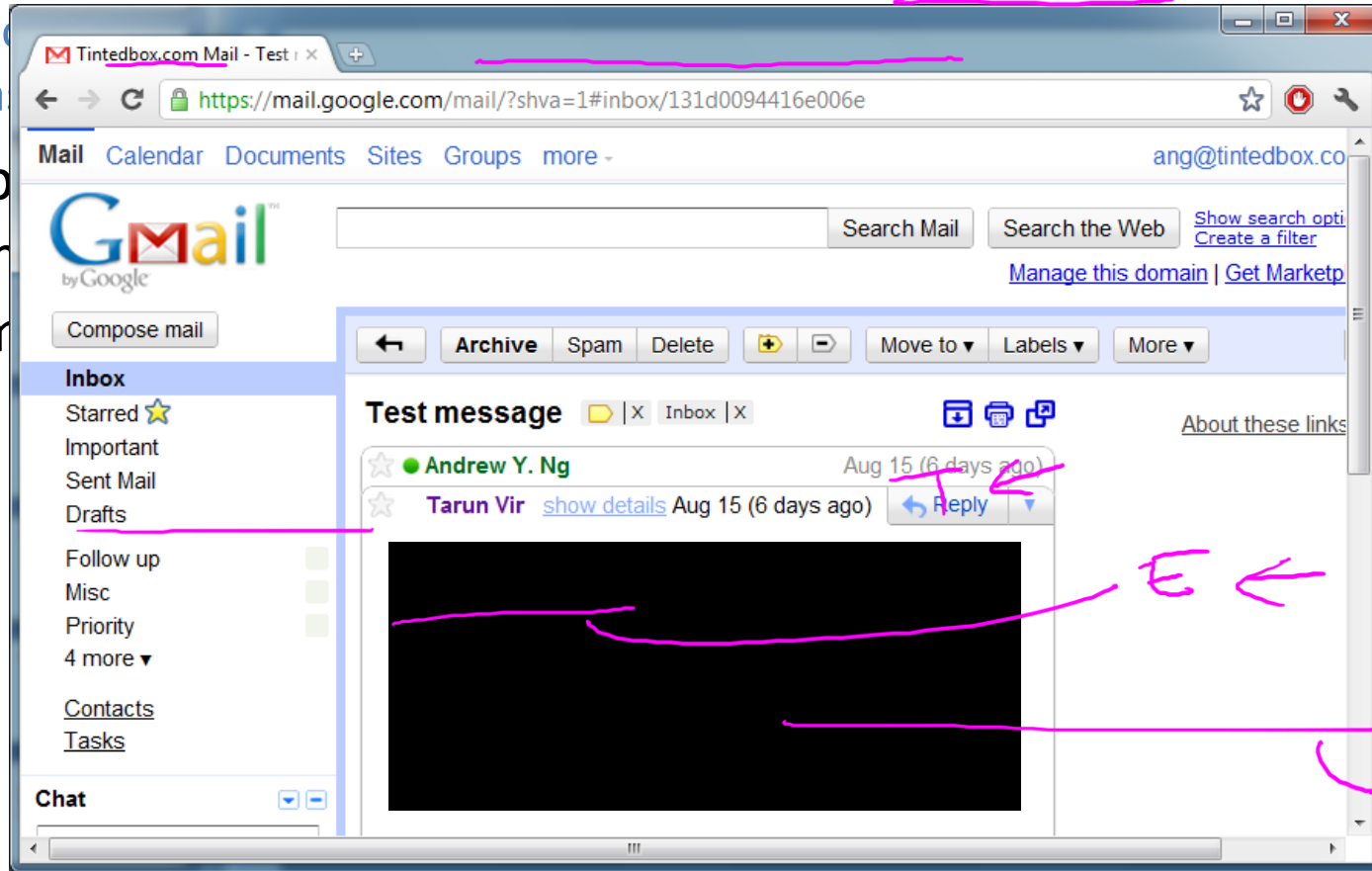
Machine Learning definition

- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

"A computer program is said to *learn* from experience E with respect to T ,

Support
not r
spam

or do
filter



spam.

P

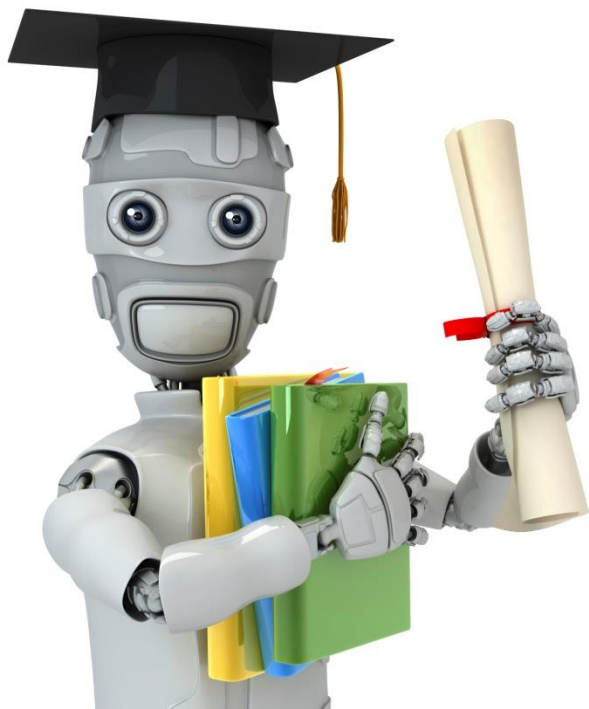
Machine learning algorithms:

- Supervised learning
 - Unsupervised learning
- 

Others: Reinforcement learning, recommender systems.

Also talk about: Practical advice for applying learning algorithms.



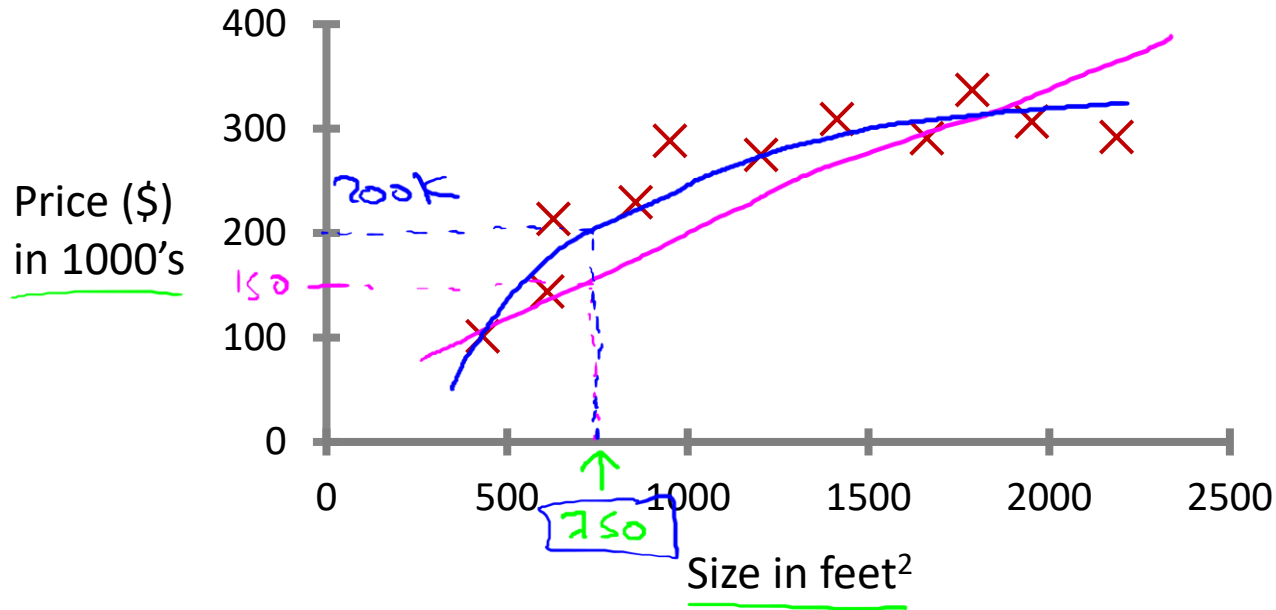


Machine Learning

Introduction

Supervised Learning

Housing price prediction.

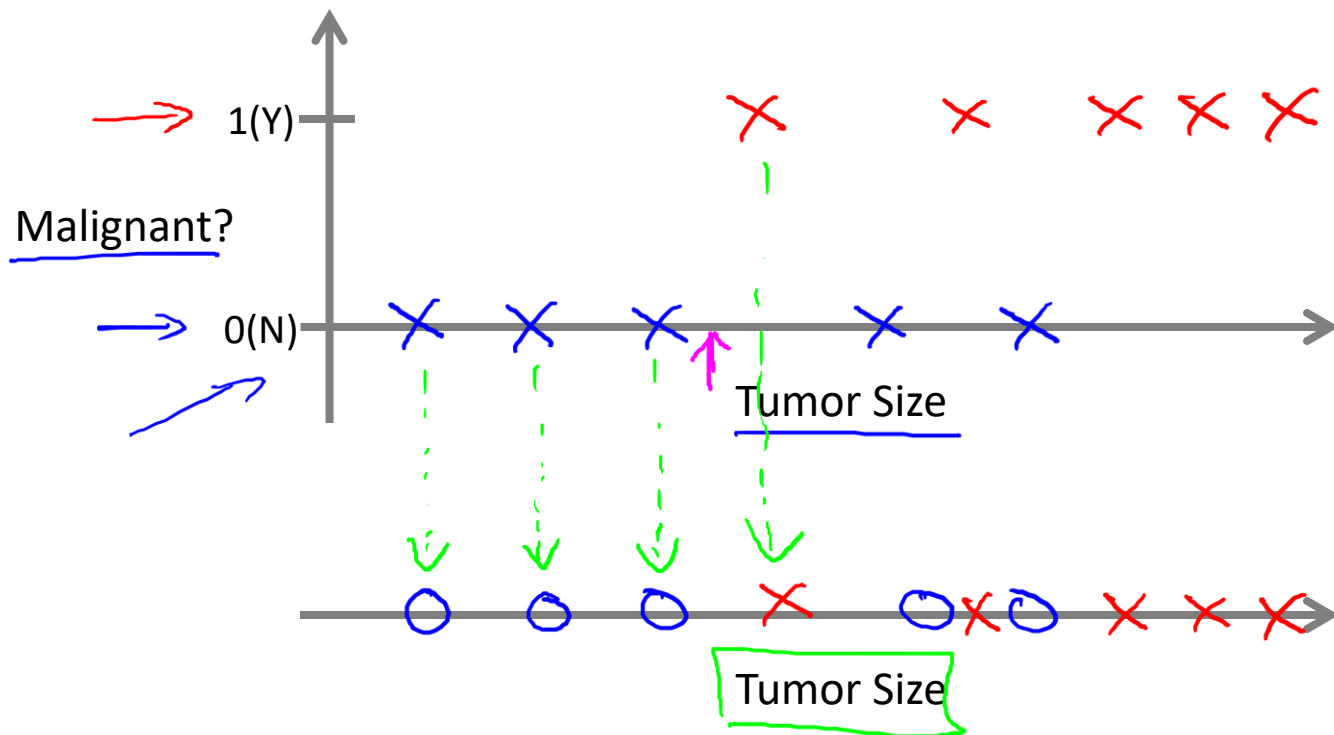


Supervised Learning

"right answers" given

Regression: Predict continuous
valued output (price)

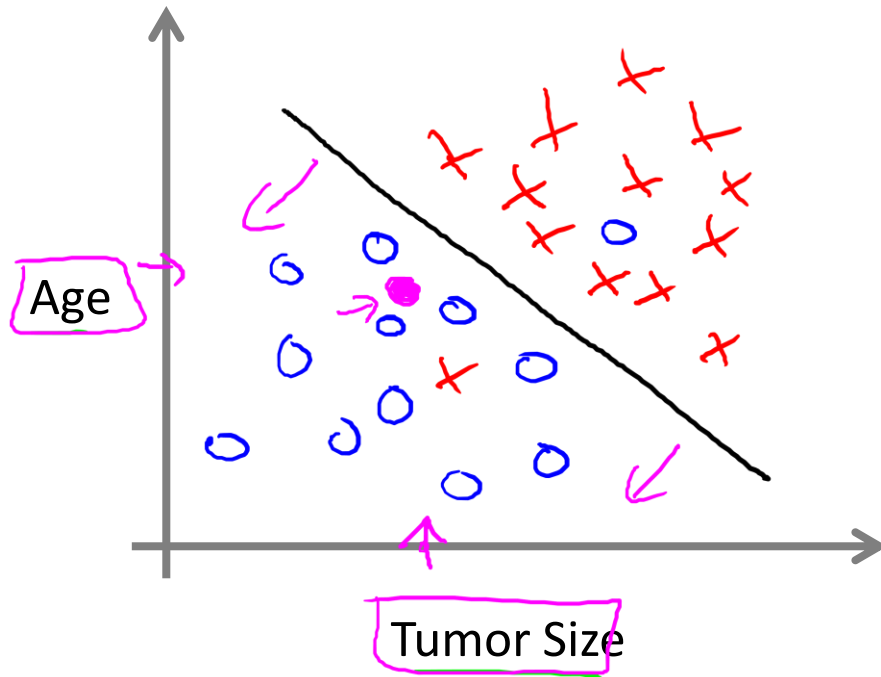
Breast cancer (malignant, benign)



Classification

Discrete valued
output (0 or 1)

0, 1, 2, 3
↓ ↓ ↓ ↓
benign type 1
cancer



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

You're running a company, and you want to develop learning algorithms to address each of two problems.

1000's

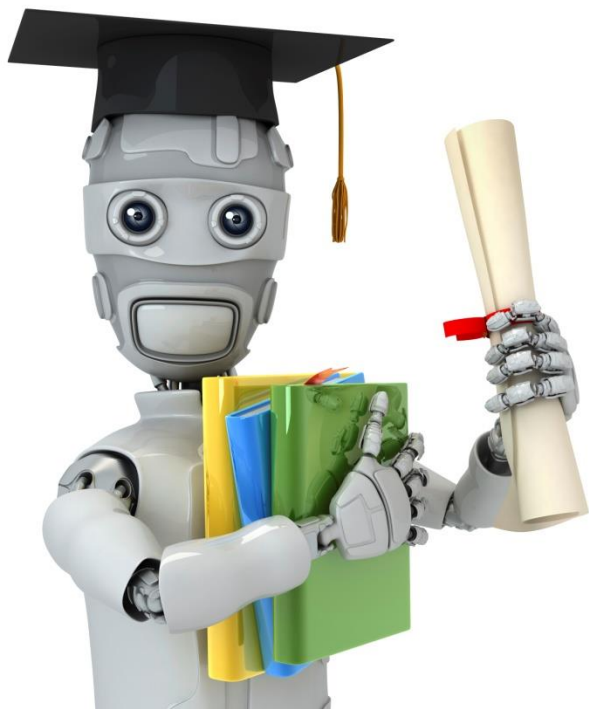
↗ Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.

↗ Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

↗ 0 - not hacked
↗ 1 - hacked

Should you treat these as classification or as regression problems?

- ☐ Treat both as classification problems.
- ☐ Treat problem 1 as a classification problem, problem 2 as a regression problem.
- ☐ Treat problem 1 as a regression problem, problem 2 as a classification problem.
- ☐ Treat both as regression problems.

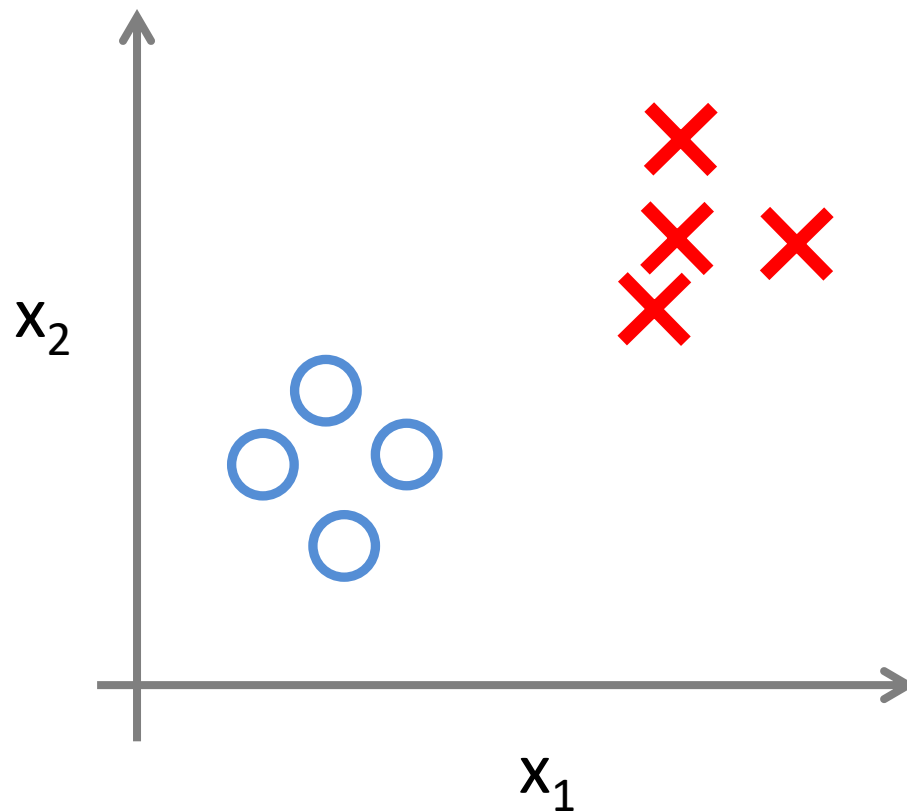


Machine Learning

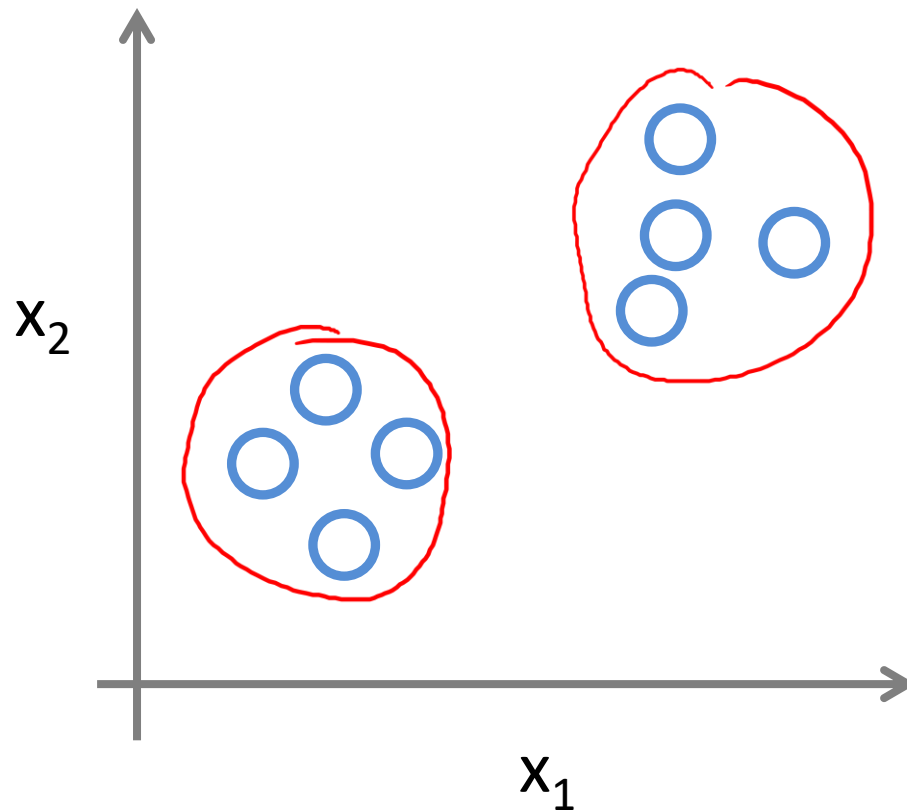
Introduction

Unsupervised Learning

Supervised Learning



Unsupervised Learning



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Allen: Well is dead, but much Gulf Coast work remains

By the CNN Wire Staff
September 20, 2010 -- Updated 1317 GMT (2117 HKT)



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What next for Gulf oil spill?

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Julia Kollewe

guardian.co.uk, Monday 20 September 2010 08.33 BST

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BP's costs for the Deepwater Horizon disaster have hit \$10bn. Photograph: Ho/Reuters



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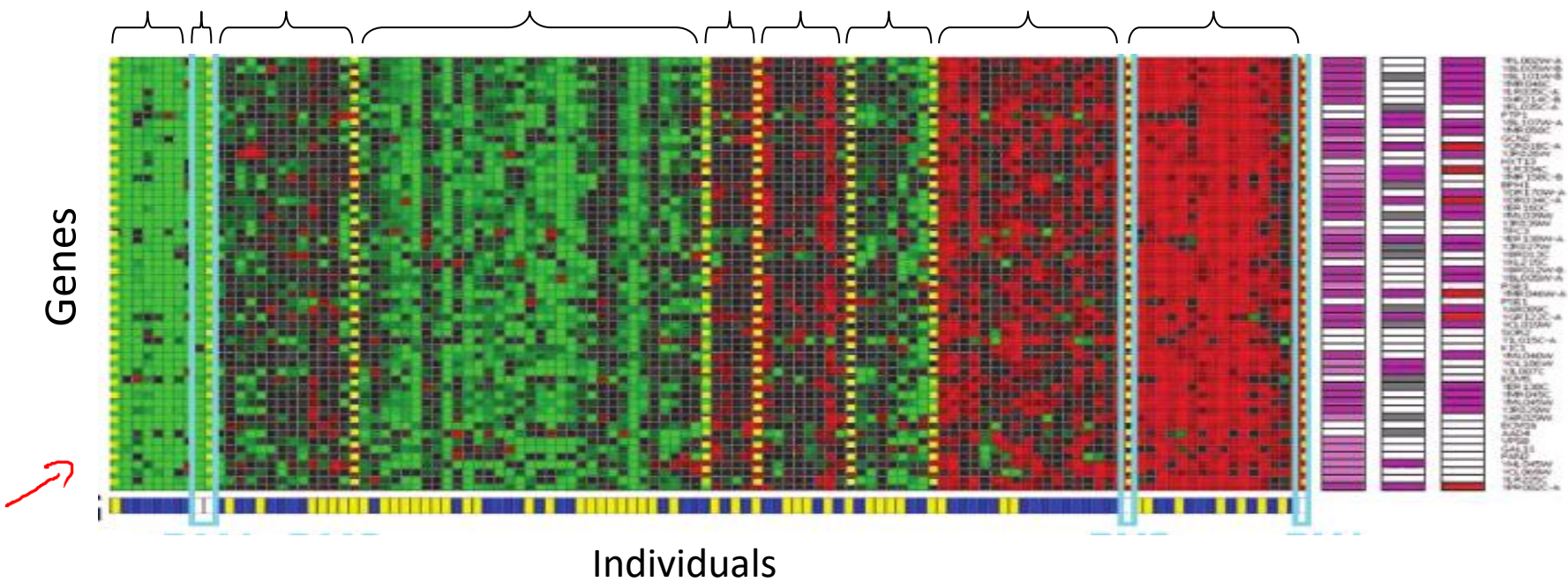
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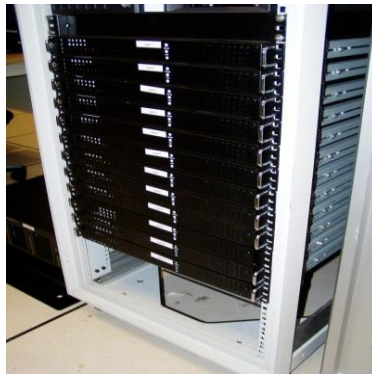
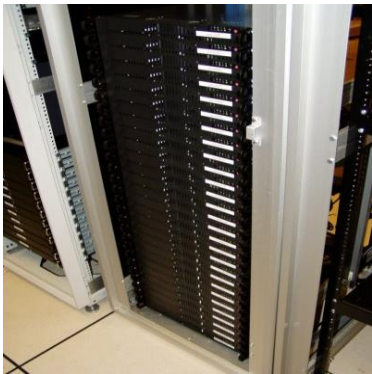
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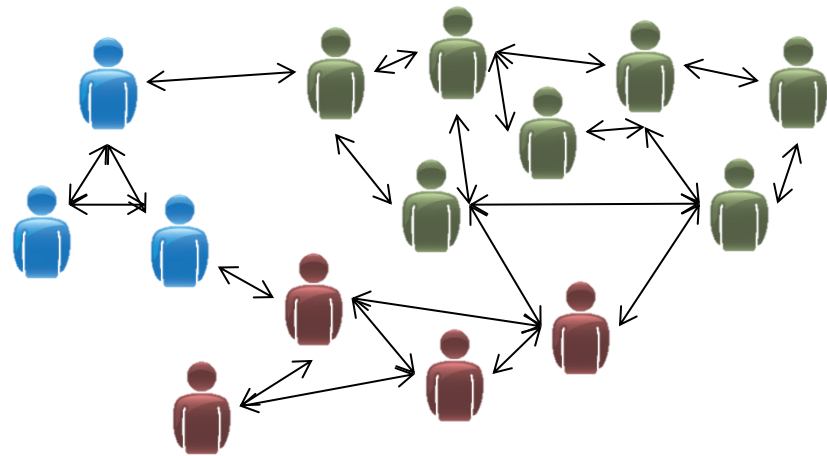
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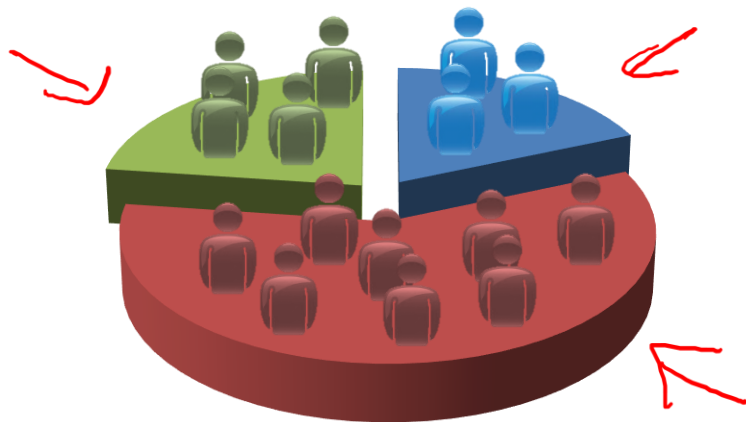
[Source: Su-In Lee, Dana Pe'er, Aimee Dudley, George Church, Daphne Koller]



Organize computing clusters



Social network analysis

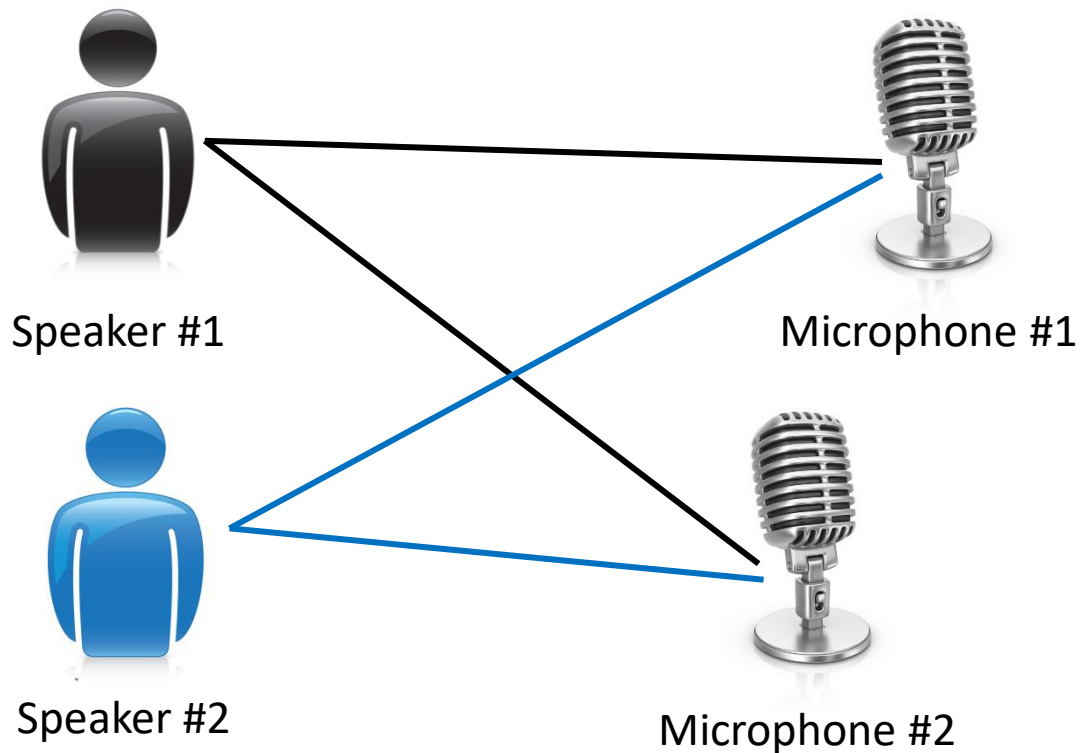


Market segmentation



Astronomical data analysis

Cocktail party problem



Microphone #1: 📢

Output #1: 📢

Microphone #2: 📢

Output #2: 📢

Microphone #1: 📢

Output #1: 📢

Microphone #2: 📢

Output #2: 📢

Cocktail party problem algorithm

```
[W,s,v] = svd(( repmat(sum(x.*x,1),size(x,1),1).*x)*x');
```

Of the following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☐ Given a set of news articles found on the web, group them into set of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

